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Levi Neal

Lyon College, Levi.neal@lyon.edu

Lauren Scott

University of Kansas, laurensconfig529@gmail.com

Gosia Zobel

EthicoNZ, gosia@ethiconz.com

See next page for additional authors

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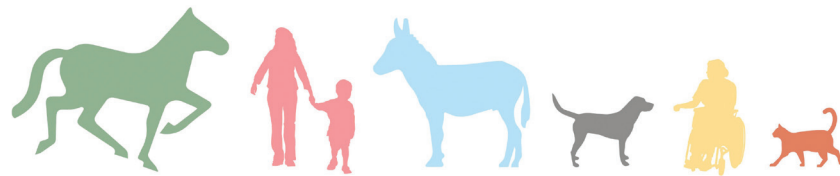
Paws and Predictions: Interpreting Intraspecific Cat Social Interactions through Facial Signals

Cover Page Footnote

We thank the staff and volunteers at the CatCafé Lounge for granting us access to their cat population and for their extensive insight on cat behavior during data collection.

Authors

Levi Neal, Lauren Scott, Gosia Zobel, and Brittany Florkiewicz



Paws and Predictions: Interpreting Intraspecific Cat Social Interactions through Facial Signals

Levi Neal,¹ Lauren Scott,² Gosia Zobel,^{3,4} and Brittany N. Florkiewicz^{5*}

Keywords: domestic cats, cat behavior, predictive accuracy, human perception, facial signals

ABSTRACT There has been limited research on humans' ability to perceive and predict social interactions among cats using facial signals. This area is crucial to explore since conflicts between cats often cause significant stress for their guardians. By evaluating individuals' skills in understanding cat interactions, researchers can develop tools to help guardians predict affiliative and nonaffiliative behaviors, leading to effective intervention strategies for multicat households. We distributed a survey to individuals subscribed to the Companion Animals New Zealand newsletter to assess cat guardians' effectiveness in perceiving and predicting various social interactions among cats. We hypothesized that guardians with one or more cats (P1), those who grew up with cats (P2), or those who frequently interacted with cats (P3) would perform similarly to those without such experiences. Additionally, we predicted that guardians with professional experience (P4) would outperform those without. The survey consisted of four parts: (1) assessing participants' experiences and knowledge of cats; (2) an experimental video paradigm; (3) a follow-up assessment; and (4) a demographic questionnaire. Participants viewed 10 video clips (five affiliative and five nonaffiliative) showcasing the beginning of cat interactions and were asked to predict the outcomes based on facial signals. Specifically, participants had to determine whether the outcome would be friendly or not and describe their predictions. Afterward, they watched complete interactions to revise their responses. We found that participants struggled most with predicting nonaffiliative interactions based on facial signals. Predictions P1 to P3 received full support, while P4 received partial support; participants with professional experience performed significantly better at identifying nonaffiliative facial signals. Although our sample of videos and facial signals is limited, our results suggest that creating training tools to

(1) Department of Biology, Lyon College, Batesville, Arkansas, USA, (2) School of Medicine, University of Kansas Medical Center, Kansas, USA, (3) Companion Animals New Zealand, Wellington, New Zealand, (4) EthicoNZ, Hamilton, New Zealand, (5) Department of Psychology, Lyon College, Batesville, Arkansas, USA

*Corresponding Author. Email for correspondence: brittany.florkiewicz@lyon.edu

help participants recognize and respond to nonaffiliative facial signals is a promising direction. Additionally, early intervention strategies may prove to be the most effective approach.

Introduction

While many domesticated cats roam freely worldwide, a significant number (approximately 200 million) are kept indoors (Bernstein, 2007; Gyles, 2019). Multicat households (Bernstein & Strack, 1996; Elzerman et al., 2020; Khoddami et al., 2023) and shelters (Gouveia et al., 2011; Kessler & Turner, 1999; Loberg & Lundmark, 2016) are increasingly common, leading to more opportunities for intraspecific social interactions. Like their free-ranging counterparts, indoor cats produce a range of both affiliative and nonaffiliative behaviors to one another on a daily basis (Elzerman et al., 2020). Affiliative behaviors can include co-sleeping, grooming, and nose-touching (Cafazzo & Natoli, 2009; Crowell-Davis et al., 2004; Dards, 1983; Elzerman et al., 2020; K. R. Vitale, 2022), while nonaffiliative behaviors consist of staring, chasing, growling, and stalking (Elzerman et al., 2020; Natoli et al., 2001; Penar & Kloczek, 2018; Stelow et al., 2016; Yeon et al., 2011). Guardians (i.e., pet owners and caretakers of shelter cats) often consider nonaffiliative behaviors problematic (Powell et al., 2023), and these behaviors can become a source of stress for guardians over time (Van Belle et al., 2023). Resources that help guardians understand their cats' behavior, anticipate the results of social interactions, and respond effectively to such interactions are crucial for minimizing guardian stress and improving the overall health and well-being of their cats. Numerous studies have addressed misconceptions that guardians have about the social behavior and care needs of domesticated cats (Croney et al., 2023; Grigg & Kogan, 2019; Menchetti et al., 2018; Natoli et al., 2022; Udell et al., 2023). However, few studies have examined whether guardians of multiple domesticated cats can accurately predict the outcomes of their cats' social interactions. Without this information, guardians may struggle to address potential conflicts effectively.

One possible reason for the gap in research regarding guardians' ability to predict cat-cat interactions

is the general lack of research on cat facial signaling behavior. Generally speaking, signalers may display facial signals as a result of their emotional experiences (Darwin, 1872; Ekman, 1970; Ekman, 2009; *What the face reveals*, 2005) and/or to achieve specific goals (such as requesting grooming or initiating a play bout; Cartmill & Byrne, 2010; Florkiewicz & Campbell, 2021; Liebal et al., 2006). Recipients can interpret these signals to predict the signaler's future behaviors that stem from those experiences/goals (Preuschoft, 2000; Preuschoft & van Hooft, 1995; Waller et al., 2017). Both cats (Quaranta et al., 2020) and humans (Fridlund, 1994) frequently use facial signals to predict the behaviors of signaling conspecifics. For example, when a cat produces a nonaffiliative facial signal, it typically flattens its ears against its head and displays its teeth and/or tongue (Bennett et al., 2017; Scott & Florkiewicz, 2023). When a conspecific or a human observes this nonaffiliative signal, they may choose to distance themselves from the signaling cat, recognizing that the signal indicates a high likelihood of aggression. Although much has been published about the facial signals that cats direct toward humans (Bennett et al., 2017; Caeiro et al., 2017; Dawson et al., 2019; Ellis, 2018), few studies in comparison have been published about the facial signals that cats direct toward each other (Brown & Bradshaw, 2014; Scott & Florkiewicz, 2023). However, two recent studies have revealed that cats produce a wide variety of facial signals (Scott & Florkiewicz, 2023), many of which can be rapidly mimicked to enhance social bonding (Martvel et al., 2024). These studies indicate that cats can create various facial signals that humans may interpret to understand the cats' emotions and imperative goals, helping to predict their future behaviors.

Human interpretations of cat facial signaling behavior have been assessed (Caeiro et al., 2017; Dawson et al., 2019; de Mouzon et al., 2024). These studies often involve perceptual rather than predictive assessments and have limitations regarding their scope and methodology. For instance, one study

investigated whether the facial signaling behavior of shelter cats could predict how quickly they were adopted (Caeiro et al., 2017). However, the facial signals were directed toward a human experimenter prior to adoption, leaving it uncertain how the adopters perceived the cats' facial signals themselves. Another study looked into whether humans can accurately categorize the facial signals of cats as being associated with either a positive or negative affective state (Dawson et al., 2019). However, the videos used in this study mainly focused on the behavior of the signaling cat at the time the signals were produced. They did not include assessment measures that would help determine whether human participants could predict the outcomes following these positive and negative facial signals. Additional studies have also been conducted on whether human perceptions of cat facial signals change when combined with vocal signals, but these studies still rely on perceptual phenomena, again focusing on the behavior of the signaling cat at the time the signals were produced (de Mouzon et al., 2024).

Research Questions

Our study aims to evaluate whether guardians—past, present, or prospective—can accurately predict the outcomes of social interactions among domesticated cats (*Felis silvestris catus*) using information about how they perceive facial signals.

Consistent with previous studies (Dawson et al., 2019; de Mouzon et al., 2024), we have made several predictions about the relationship between guardian demographics and accuracy—both perceived and real—in predicting the outcomes of intraspecific cat social interactions. Previous studies indicate that cat guardianship, both past and present, had minimal impact on participants' ability to accurately categorize the affective state (i.e., underlying emotional condition) of cat facial signals (Dawson et al., 2019). We predict that guardians with one or more cats (P1), who grew up with cats in their childhood homes (P2), and/or who frequently interacted with cats while growing up (P3) will perform similarly to those

without these experiences. However, previous studies have found that having professional experience with animals increases the ability to accurately classify the affective state (or emotional condition) of cat facial signals. We anticipate that guardians in our study will also predict the outcomes of intraspecific cat social interactions more accurately if they have professional animal experience (P4; Dawson et al., 2019). It is important to note that while our study predictions align with Dawson et al.'s 2019 study, we are situating these within both perceptual and predictive frameworks, rather than perceptual frameworks alone.

Methods

Our research was conducted at the CatCafé Lounge, a nonprofit rescue organization located in Los Angeles, CA, USA, accommodating up to 30 adult cats available for adoption at a time. The dynamic nature of adoptions meant that our sample size extended beyond 30 individuals. Cats had access to indoor and outdoor seating areas equipped with cat beds, toys, scratching posts, and hiding spots. Cats had access to a quiet room via one of multiple cat doors. This room is not accessible to visitors and offers unlimited food, water, and litter boxes. Visitors are allowed to enter the café by appointment (between 11:30 a.m. and 5:30 p.m.). Daily cleaning is performed by various caregivers before and after visiting hours. Our study was approved by the CatCafé Lounge and was conducted in accordance with the Association for the Study of Animal Behaviour's *Guidelines for the Treatment of Animals in Behavioral Research* (2020). Full IACUC approval was waived for our study, as we utilized noninvasive behavioral observations (i.e., video recordings collected in regular visitor areas without interacting with or altering the routines of the cats).

Cat Video Footage

Video footage of intraspecific cat social interactions was collected by the second author (LS) over a 150-hour period using an opportunistic sampling method

(Florkiewicz & Campbell, 2021; Scott & Florkiewicz, 2023) from August 2021 to June 2022. We followed the largest and most active group of cats, largely in the indoor and outdoor seating areas, and only outside of visitor hours to reduce the possibility of capturing cat-human interactions and to limit human interference. Recordings were initiated right before a social interaction began (using the pre-record function on the Panasonic Full HD Video Camera Camcorder HC-V770) and ended as soon as the cats stopped communicating or dispersed. A total of 168 video clips captured facial signaling behaviors during 186 social interactions from 53 adult (1 yr +) domestic short-haired cats.

Cat Video Coding

The onset and offset of all intraspecific cat social interactions were marked, and then all observed facial signals were coded. A facial signal was defined as facial muscle movements that are produced to influence the behavior of conspecifics (Florkiewicz et al., 2023; Scott & Florkiewicz, 2023; Smith & Harper, 1995). We excluded facial muscle movements essential for biological functions, such as breathing and chewing, as well as head and eye movements, because it was challenging to determine if they served a communicative purpose (Florkiewicz et al., 2023). We only coded facial muscle movements and the corresponding signals directed toward conspecifics; we omitted signals directed toward humans and inanimate objects. For each facial signal, we coded the identity of both the signaler and the recipient. All facial signals produced by both signalers and recipients, along with their associated social interactions, were categorized as either affiliative or nonaffiliative based on the accompanying behaviors immediately after signal production. Affiliative behaviors included allogrooming, rubbing, body touching, nose touching, social rolling, mating, play, and/or vertical tail positioning (Crowell-Davis et al., 2004; K. Vitale, 2022; K. R. Vitale, 2022). Nonaffiliative behaviors included biting, growling, fleeing, hissing, scratching, piloerection, spitting, swatting, and/or staring (Penar & Kloczek, 2018; Stelow et al., 2016).

All individual facial muscle movements (i.e., action units—AUs) produced at the apex (or peak) of each facial signal (Florkiewicz et al., 2018) were coded using the Facial Action Coding System (or catFACS) (Caeiro et al., 2017). The primary coder for all video clips was the last author (BNF), with 10% of video clips being reassessed by the second author (LS) for interobserver reliability. Specifically, we assessed the context of facial signals and social interactions by using a percentage of agreement (Florkiewicz & Campbell, 2021) and FACS coding with Wexler's ratio (Caeiro et al., 2017). Wexler's ratio is a metric used to assess the level of agreement between two established catFACS coders. It is calculated by taking the number of facial muscle movements (or AUs) that both coders agreed on, multiplying that number by 2, and then dividing it by the total number of AUs scored by both coders, which includes both agreed and disagreed movements (Wathan et al., 2015). Wexler's ratio is particularly useful for evaluating FACS agreement, given the vast number of potential codes and the large volume of items that coders may score (Wathan et al., 2015). We achieved good agreement for both contexts (75%) and FACS coding (0.707).

Video Selection Process

The 168 video clips collected by Scott and Florkiewicz (2023) were evaluated to determine their suitability according to the criteria that reduce ambiguity and ensure accurate responses from participants. The first 4 criteria yielded 40 suitable video clips, consisting of 28 affiliative clips and 12 nonaffiliative clips. Applying the additional 3 criteria allowed us to decrease our pool of suitable clips from 40 to 10, consisting of 5 affiliative clips and 5 nonaffiliative clips.

1. The video showcased a social interaction that was clearly either affiliative or nonaffiliative, including only behaviors distinct to that category. Interactions that included behaviors from both categories, affiliative and nonaffiliative, or behaviors not listed were excluded as viable options.

2. The video featured a single social interaction between one signaler and one recipient to minimize the inaccuracies associated with identifying key facial muscle movements in multiple cats (Molina et al., 2019).
3. The video must clearly show the beginning, duration, and end of the social interaction. It should include at least 3 seconds of footage before and after the interaction occurs to ensure that participants can view a complete and isolated interaction.
4. The first facial signal produced by signaler (i.e., the cat initiating the interaction) should not overlap with any facial signals produced by other cats, and any further facial signals should occur only after the initial signal has finished. There should be a minimum of 1 second between the end of the signaler's first facial signal and the start of any behavioral responses from the recipient.
5. Are the cats (signaler and recipient) clearly visible? Is the video clip focused on these two cats, or are there distractions occurring in the background?
6. Are the facial muscle movements of each cat clearly visible?
7. Were the behaviors associated with each context, affiliative and nonaffiliative, clear and easy to recognize?

Video Editing Process

Each of the 5 affiliative and 5 nonaffiliative video clips was trimmed to include 3 s before and after each intraspecific social interaction occurred, and two versions were created: (1) a partial clip video, which includes the 3 s leading up to the first facial signal and ends shortly after this signal (to capture the participant's prediction of the outcome), and (2) a full clip video, which includes the entire intraspecific social interaction from start to finish (to capture the participant's perception of the actual outcome). A green arrow pointing to the cat producing the first facial signal of the social interaction was added. Each video (available in the **electronic supplement**)

was assigned a unique numerical code along with a letter (A or B) to indicate the type of clip it represented, distinguishing between partial and full clips.

Human Participants

The study was approved by Lyon College's Institutional Review Board (IRB; Approval Code: FA24-05-092324; **electronic supplement**). Participants were recruited through the newsletter issued by Companion Animals New Zealand (CANZ). CANZ is a New Zealand-based animal welfare charity that currently has approximately 200,000 people voluntarily signed up to the organization's newsletter. Potential participants could click through to a SurveyMonkey link directly from the newsletter. A total of 598 people participated in the survey. The median age of our participants was 53 years (range = 18–84). Most (85%) of the participants were female ($n = 510$) and of European descent ($n = 483$ or 81%).

Survey Measures

The survey (see **electronic supplement**) was divided into four parts.

Part 1. Participants were asked a series of Likert-scale questions (1 = strongly disagree, 5 = strongly agree) regarding their exposure to, and experience with, cats and other animals. These questions were utilized to test predictions P2 and P3 (**Table 1**). Participants were also asked additional questions about how good they are at reading the body language of cats and other animals, and how good they are at predicting the behaviors of these animals (see **electronic supplement**).

Part 2. Participants were informed that they would watch 10 sets of videos, each consisting of one partial clip and one full clip. No limit was placed on how many times participants could watch the video clips; however, once they clicked the “next” button, they were not permitted to return to the previous video clips or questions to reduce the risk of participants changing their answers. While watching

Table 1. The survey questions that we used to test our four research predictions (P1–P4).

Question	Type	Prediction
Do you currently share your household with one or more cats?	Binary	P1: Guardians who currently have one or more cats will exhibit similar performance to those who do not have cats.
Growing up, my household had many cats.	Likert-scale	P2: Guardians who grew up in homes with cats will perform similarly to those who did not grow up with cats.
Growing up, I had frequent interactions with cats.	Likert-scale	P3: Guardians who frequently interacted with cats during their childhood will show similar performance to those who did not have frequent interactions with cats.
Do you currently, or have you in the last 5 years, had a job or occupation that involved working directly with animals?	Binary	P4: Individuals with professional experience in working with animals will perform significantly better at predicting interactions between cats compared to those without professional animal experience.

partial clips, participants were told to focus on the signaler’s face (indicated by the green arrow). They were then asked to predict if the social interaction would be friendly (i.e., affiliative; yes/no answer), and also to briefly describe how they thought the recipient cat would respond to the signaling cat’s behavior. Viewers were then directed to the full video clip; after seeing the entire interaction, they were asked if they felt the social interaction was friendly and also how they felt the recipient cat responded. The order of video clip sets was randomized for each participant; however, the partial clip was always shown before its corresponding full clip. By asking participants to respond to questions regarding the context and behavioral outcomes for both the partial and full clip conditions, we aimed to evaluate their confidence in perceiving and predicting behavior.

Part 3. To contextualize the results and guide the development of effective teaching and training tools for the future, participants were asked about their animal body language and behavior predicting abilities again (available in the **electronic supplement**).

Part 4 Participants were asked demographic questions, including their occupational experiences with animals and how many animals they currently care for (including cats). These questions were utilized to test predictions P1 and P4 (**Table 1**).

Data Handling and Analysis

Participant survey data were gathered into an Excel sheet and imported directly into R 4.1.1 (Chambers, 2008). A copy of our raw data and R code can be found in the **electronic supplement**.

The participants’ answers to Part 3 were compared to the coded responses of BNF and LS, who have extensively researched mammalian facial signaling and have prior experience working with cat facial signals (Mahmoud et al., 2025; Martvel et al., 2024; Scott & Florkiewicz, 2023). If the participant identified the same context as the researcher, their response was coded as 1. Conversely, if they identified a context opposite to that of the researcher, their response was coded as 0. The highest score a participant could achieve was 5, as each participant watched 5 affiliative and 5 nonaffiliative videos for both the partial and full conditions. This resulted in a total of 4 scores per participant (affiliative partial, nonaffiliative partial, affiliative full, and nonaffiliative full). A score of 0 indicates no agreement between BNF/LS and the participant for the video condition, while a score of 5 signifies perfect agreement.

To assess whether participant scores improved after watching the full video clip, or if participants continued to struggle with the footage, each participant’s scores were compared using paired *t*-tests. To evaluate whether the participants’ confidence in

understanding and predicting the outcomes of social interactions increased or decreased, their responses to the Likert-scale questions administered before and after the videos were compared using Wilcoxon Signed Rank Tests, based on the average Likert-scale score for each question type. We assessed whether participants were significantly more likely to change their responses between the partial and full video clip conditions using a McNemar Test for each video set.

To gain a deeper understanding of whether there are significant differences in participant scores based on their childhood experiences with cats (predictions P2 and P3), the number of cats they currently care for (P1), and whether they have had an animal-related occupation in the past five years (P4), ordinal regressions were used and natural language processing was implemented. We also used ordinal regression models to evaluate whether participants' confidence in reading cat body language and predicting their behavior, prior to the experimental portion of our survey, influenced their responses during the partial clip condition.

Ordinal Regression

The participant scores for the partial clip condition, which has a natural score ordering from 0 to 5, were set as the dependent variable. Modeling was conducted separately for both affiliative and non-affiliative social interactions to determine whether different factors influence performance in interpreting these contexts. For each context (affiliative and nonaffiliative), we ran three ordinal regression models. The first model included independent variables with present-day information (number of cats in the household and animal-related occupation) used to test predictions P1 and P4 (**Table 1**). The second model included independent variables with childhood experiences with cats (whether the participants' childhood home contained cats and whether they had frequent interactions with cats) used to test predictions P2 and P3 (**Table 1**). The final model included questions about participants' confidence in assessing the body language of cats and predicting their behavior. All models were run using the "ordinal" package in R (Christensen, 2019). Due to a bias

in our participant pool toward a specific age and sex group, these random effects were excluded.

Natural Language Processing

To capture the specific features of the facial signals of cats that participants focus on when making their decisions, as well as other features that may influence their scores (e.g., positional behaviors, fur coloration), natural language processing (NLP) was used. First, we combined the written responses of participants to the partial clip condition for affiliative and nonaffiliative intraspecific social interactions. Next, participant responses were filtered for punctuation, spacing, and irrelevant articles (such as "the") using the "quanteda" (Benoit et al., 2018), "stringr" (Wickham & Wickham, 2019), and "tm" (Feinerer et al., 2015) packages. Using the "tm" package (Feinerer et al., 2015), frequency data for individual words and co-occurrence data for groups of words were extracted, and the top 25 words were selected. We also analyzed common word pairings and associations using the "tm" package (Feinerer et al., 2015) to better understand the relationships between key terms. We identified the top pairings for both affiliative and nonaffiliative social interactions. Pairings are presented below if there are 10 or more co-occurrences. We excluded word pairings related to the video setup, such as "cat" with "arrow" and "green," as they often appeared with other terms and were less insightful for our analyses.

Results

Participant Demographic Information

The majority of our 598 participants were guardians of at least one cat ($N = 519$, or 87%). Over half of the participants ($n = 337$ or 56%) attended university. Most participants did not currently have children living with them ($n = 494$ or 83%) and resided in detached homes ($n = 453$ or 75.80%). Most participants frequently interacted with cats during their childhood (Likert scale average = 4.33, SD = 1.17) and previously had cats as pets (Likert scale average = 3.58, SD = 1.46). Participants showed a strong

preference for interacting with their own cats (Likert scale average = 4.94, SD = 0.32), but enjoyed interacting with other cats (Likert scale average = 4.82, SD = 0.45), as well as different types of companion animals besides cats (Likert scale average = 4.44, SD = 0.82), and livestock and wildlife (Likert scale average = 4.13, SD = 0.99).

Identifying Affiliative and Nonaffiliative Interactions

There was no difference between participants' performance for affiliative cat social interactions ($t = -1.838, p = 0.067$), with an average Likert scale score

of 4.418 (SD = 0.808) for the partial clips and 4.480 (SD = 0.907) for the full clips. However, for nonaffiliative cat social interactions, participants did better ($t = -22.661, p < 0.001$) when observing the full clip (average = 4.341, SD = 0.790) compared to the partial clip (average = 3.485, SD = 0.936).

Experience with Animals and Accuracy of Predictions

For affiliative interactions, there were no differences in participant scores based on any of our independent variables of interest (**Table 2**). However, participants currently or recently (i.e., within the

TABLE 2. Effects of participants' current-day experiences, childhood experiences, and confidence in their ability to read cat body language and predict cat behavior prior to participating in the experiment on their scores when identifying affiliative and nonaffiliative social interactions.

Model ID	Independent Variable	β	SE	z-value	p-value
Affiliative_Present	(Threshold Coefficients)	Variable	Variable	Variable	N/A
	Cats in Current Household	0.102	0.2368	0.432	0.666
	Animal-Related Occupation	0.183	0.241	0.758	0.448
Nonaffiliative_Present	(Threshold Coefficients)	Variable	Variable	Variable	N/A
	Cats in Current Household	-0.172	0.220	-0.783	0.434
	Animal-Related Occupation	0.468	0.223	2.10	0.036
Affiliative_Childhood	(Threshold Coefficients)	Variable	Variable	Variable	N/A
	Cats in Childhood Home	0.065	0.080	0.817	0.414
	Frequent Interactions	0.002	0.098	0.022	0.982
Nonaffiliative_Childhood	(Threshold Coefficients)	Variable	Variable	Variable	N/A
	Cats in Childhood Home	0.051	0.073	0.697	0.486
	Frequent Interactions	-0.013	0.093	-0.138	0.890
Affiliative_Confidence	(Threshold Coefficients)	Variable	Variable	Variable	N/A
	Reading Cat Body Language	0.1346	0.174	0.773	0.439
	Predicting Cat Behavior	0.016	0.1522	0.108	0.914
Nonaffiliative_Confidence	(Threshold Coefficients)	Variable	Variable	Variable	N/A
	Reading Cat Body Language	0.048	0.159	0.304	0.761
	Predicting Cat Behavior	0.234	0.141	1.663	0.096

last 5 years) involved in animal-related occupations scored higher in nonaffiliative social interactions compared to those without such experience ($\beta = 0.468$, $SE = 0.223$, $z\text{-value} = 2.10$, $p\text{-value} = 0.036$). No differences were found for any other factors regarding nonaffiliative scores.

Potential Sources of Disagreement

A total of 34 unique terms were used to describe affiliative and nonaffiliative partial clips, and 16 were found across both contexts (affiliative and nonaffiliative) (**Table 3**). The terms “play” and “fight” were the most frequently mentioned pairing in our participants’ responses to nonaffiliative social interactions in the partial clip condition (**Table 4**). Notably, the co-occurrence of word pairs in nonaffiliative responses is relatively low compared to that in affiliative responses. The average number of words was higher for nonaffiliative social interactions (average = 3.92, $SD = 3.26$) than affiliative social interactions (average = 3.56, $SD = 2.92$).

Participant Confidence Before and After the Paradigm

In the post-test survey, participants’ answers changed in 5 of the 8 questions aimed at assessing their confidence in reading the body language of animals, including cats, and predicting their behavior (**Table 5**). Participant confidence in reading the body language and predicting the behavior of cats other than their own decreased. Participants also showed a decreased confidence in reading body language and predicting the behavior of other companion animals in the post-test condition.

Finally, we examined whether participants were more or less likely to change their answers after viewing the full clip condition. Out of the 10 videos (5 affiliative and 5 nonaffiliative), participants were more likely to change their responses for 6 of them (McNemar Test; $p < 0.001$). Of these six videos, four were associated with nonaffiliative interactions. For the four nonaffiliative video clips, three of these clips were associated with changes between the partial

TABLE 3. The top 25 words identified through frequency counts for partial video clips, separated based on the context of the video (affiliative versus nonaffiliative). There were 34 unique words, and those that are italicized appear in the top 25 list for both contexts.

Ranking	Affiliative	Nonaffiliative
1	<i>Cat</i>	<i>Cat</i>
2	<i>Play</i>	<i>Arrow</i>
3	<i>Arrow</i>	<i>Back</i>
4	<i>Fight</i>	<i>Green</i>
5	<i>Green</i>	<i>Run</i>
6	<i>Jump</i>	<i>Attack</i>
7	<i>Friendly</i>	<i>Jump</i>
8	Playful	<i>Play</i>
9	Sniff	Black
10	<i>Back</i>	<i>Fight</i>
11	Roll	Treat
12	Playfully	<i>Swipe</i>
13	<i>Run</i>	<i>Walk</i>
14	<i>Cats</i>	<i>Friendly</i>
15	<i>Attack</i>	<i>Chase</i>
16	<i>Chase</i>	Ears
17	Playing	Move
18	Pounce	Continue
19	Engage	Hiss
20	Curious	Space
21	<i>Possibly</i>	Retreat
22	<i>Swipe</i>	<i>Paw</i>
23	<i>Walk</i>	<i>Cats</i>
24	<i>Paw</i>	<i>Possibly</i>
25	Fighting	Aggressive

TABLE 4. The top word pairings (i.e., word pairings with 10 or more observations) identified through frequency counts for partial video clips of affiliative and nonaffiliative social interactions.

Context	Word 1	Word 2	Words 1 + 2	Word 1 Only	Word 2 Only
Affiliative	play	fighting	35	735	63
Affiliative	play	roll	27	743	123
Affiliative	play	jump	25	745	198
Affiliative	touch	noses	24	46	74
Affiliative	play	chase	16	754	69
Affiliative	fight	back	15	377	184
Affiliative	play	friendly	12	758	232
Affiliative	play	back	10	760	189
Affiliative	interaction	friendly	10	81	234
Affiliative	friendly	manner	10	234	55
Nonaffiliative	play	fight	19	138	177
Nonaffiliative	back	fight	17	343	179
Nonaffiliative	back	ears	15	345	136
Nonaffiliative	fight	run	10	186	117

TABLE 5. Average pre- and post-test scores for Likert-scale questions regarding participants' confidence in reading body language and predicting the behaviors of various domesticated animals are presented.

Question	Avg. Score (Pre)	Avg. Score (Post)	Wilcoxon Signed Rank Test
I am good at reading the body language of my own cat(s).	4.60	4.60	$z = 0.306; p > 0.05$
I am good at reading the body language of cats in general.	4.18	3.92	$z = -7.721; p < 0.001$
I am good at reading the body language of companion animals other than cats.	3.77	3.65	$z = -4.404; p < 0.001$
I am good at reading the body language of other animals, like livestock and/or wildlife.	3.24	3.10	$z = -5.409; p < 0.001$
I am good at predicting the behavior(s) of my own cat(s).	4.51	4.56	$z = -1.451; p > 0.05$
I am good at predicting the behavior(s) of cats in general.	3.93	3.81	$z = -3.436; p < 0.001$
I am good at predicting the behavior(s) of companion animals other than cats.	3.54	3.47	$z = -2.229; p = 0.01$
I am good at predicting the behavior(s) of other animals, like livestock and/or wildlife.	2.99	2.98	$z = 0.180; p > 0.05$

and full conditions that resulted in greater agreement with the researchers.

Discussion

The goal of our study was to assess how well guardians can predict intraspecific cat social interactions by using the facial signals produced by the initiating cat. We examined four key predictions regarding how childhood and adult experiences with cats affect guardians' abilities to predict cat social interactions. Previous research has shown that having experience as a cat guardian, whether in the past or present, does not affect individuals' skills in accurately identifying the emotional context of cat facial expressions (Dawson et al., 2019; de Mouzon et al., 2024). Our results supported these predictions. Having cats currently and having grown up with cats in childhood did not impact participants' accuracy in perceiving cat facial signals or predicting the outcomes of social interactions between cats. Furthermore, based on previous studies (Dawson et al., 2019), we also predicted that guardians with professional animal experience would perform better than those without such experience. Our results provided partial support for this prediction: Professional animal experience improved perception of nonaffiliative cat facial signals and prediction of subsequent outcomes.

Natural language processing provides an explanation for why individuals with professional animal experience perform better than those without such experience in perceiving and predicting nonaffiliative facial signals and subsequent behaviors. When compiling a list of commonly mentioned words and phrases in the partial clip condition, we found that terms related to play fighting (such as "fight" and "chase") frequently appeared in participant responses for both affiliative and nonaffiliative social interactions. Certain types of high-intensity play fighting, such as rough-and-tumble play, may seem similar to actual fighting, but there are subtle interactional components that differentiate the two (such as piloerection during actual fighting; Marley et al., 2022; Smith, 1997). Of the 5 full affiliative video

clips shown to participants, 4 included social behaviors resulting from play, including rough-and-tumble play and chase play. Only one full video clip was linked to affiliative behavior, which involved initiating bodily contact but did not lead to play. Out of the 5 full nonaffiliative video clips shown to participants, 3 featured behaviors that can also occur during play, such as fleeing. However, the accompanying behaviors in the full versions of these clips included hissing, swatting, and piloerection, which are clear indicators of nonaffiliative behavior in cats (K. Vitale, 2022; K. R. Vitale, 2022). It is plausible that individuals without relevant professional experience may struggle more to differentiate between playful behavior and actual fighting among cats. While there is evidence for homologous facial signals and other communicative behaviors across species (Darwin, 1872), differences in underlying musculature, muscle movements, and their combinations exist (Mahmoud et al., 2025; Waller et al., 2020; Waller & Micheletta, 2013), likely due to the variation in socio-ecologies among species (Florkiewicz et al., 2023). One possible interpretation is that individuals without relevant experience studying the anatomy and physiology of cats are more likely to focus on the similarities between cats and humans, rather than on the key differences (Wan et al., 2012). Anthropomorphism is common in perceptions of domesticated animal behavior, often leading to misrepresentations of their internal mental states and inaccuracies in predicting behavior (Bradshaw & Casey, 2007).

It's important to note that participants in our study only observed accompanying affiliative and nonaffiliative behaviors in the full video conditions. Partial condition clips only displayed the facial signals from the cat initiating the social interaction. It seems that participants were able to correctly identify these accompanying behaviors, as they appropriately adjusted their responses for 3 out of the 4 nonaffiliative full videos that showed changes. It appears that most difficulties in accurately predicting the outcome of nonaffiliative video clips are associated with the facial signals themselves. The identification of instances and properties of cat play bouts using communicative information is complex, as

there are multiple contextual, emotional, and motivational factors to consider (Gajdoš Kmecová et al., 2021). For instance, a signaling cat might indicate a desire to play, but the other cat's body language might indicate their desire to engage in play is low, leading to potential conflict (Gajdoš Kmecová et al., 2021). In our current study, we asked participants to pay close attention to the facial signaling behavior of the cat initiating the social interaction in the partial clip condition. However, the signaling cats in our study did not exhibit the desire to play during these specific interactions, as they clearly displayed facial signals indicating a greater likelihood of nonaffiliative behaviors. Facial signals among cats have subtle differences in physical form and social function. For instance, playful facial signals are linked to specific muscle movements in the mouth, such as opening the mouth and relaxing the jaw, which differ from other types of feline facial signals, both friendly and unfriendly (Martvel et al., 2024; Scott & Florkiewicz, 2023). Additionally, affiliative facial signals often include specific ear movements, such as rotating the ears forward instead of backward (Martvel et al., 2024; Scott & Florkiewicz, 2023). If we had asked participants to consider the recipient cat's behavior earlier in this dyadic exchange, their accuracy would have likely improved. The outcomes of social interactions depend on the behaviors of both the signaler and the respondent.

We found that much of the confusion arose from the specific characteristics of the facial signals themselves, as facial feature words like "ears" and "hiss" were prevalent in participant responses in the nonaffiliative partial clip condition. Although hiss and play face signals produced by cats share similarities in physical morphology, such as mouth opening and the exposure of the bottom row of teeth, there are important differences between the two (Bennett et al., 2017; Scott & Florkiewicz, 2023). Hissing involves the exposure of the upper teeth and the stretching of the jaw that are seldom present in a play face (Bennett et al., 2017). Interestingly, hisses and their associated facial muscle movements were absent from our nonaffiliative video clips; instead, we featured other clear indicators of nonaffiliative behavior (consistent with our previous

study; Scott & Florkiewicz, 2023), such as ear flattening. Participants seemed to believe that subsequent video clips would include hisses, based on the types of ear movements displayed. Participants may find it particularly challenging to discern the social function of ear movements because these movements do not correspond to facial muscle movements that humans can produce. If participants focus on the similarities rather than the differences between cat and human facial muscle movements, due to the tendency to anthropomorphize the facial signals of domesticated animals (Bradshaw & Casey, 2007), they may struggle to identify the social function of ear movements.

An alternative explanation for the decreased accuracy surrounding nonaffiliative facial signals may be related to the characteristics of the cats, such as their coloration. For example, the term "black" frequently appeared in participant responses after watching the nonaffiliative partial clip condition. It is plausible that certain facial muscle movements are more difficult to discern due to the lack of contrast between the fur and key regions of the face. However, the co-occurrence of the term "black" with other words only occurred once. Additional follow-up studies will need to be conducted to explore this idea further. Furthermore, participants appear to be able to identify key facial muscle movements associated with friendlier interactions even for black cats. Their scores were higher when assessing affiliative video clips in the partial condition, and they rarely mentioned movements of the ears and mouth. Instead, they were more inclined to refer to key terms related to behaviors that typically followed the production of the facial signal, such as "jump," "pounce," and "roll."

The participants' decline in confidence for reading cat body language was only observed with cats other than their own. We asked participants to evaluate the behaviors of unfamiliar cats that they had not interacted with before. However, it is essential to determine whether guardians can accurately identify nonaffiliative facial signals among their own cats and predict their subsequent behaviors. Previous studies have found that familiarity with guardians does not influence the social behaviors of cats (Vitale & Udell, 2019). These results could suggest that cats lack

idiosyncratic behaviors that could help guardians distinguish between friendly and unfriendly interactions when they are present. But it is possible that familiarity with their cats' general communicative habits and facial morphology could mediate this relationship. Even if research shows that guardians are better at understanding their own cats' facial signals and predicting their behavior, training and educational tools will still be beneficial. Guardians may still observe social interactions among cats in their daily lives, considering how common free-ranging cats are. One previous study found that young children perform better at recognizing and labeling human facial signals compared to those of dogs (Correia-Caeiro et al., 2023). The researchers concluded that while casual social interactions may be sufficient for learning about dog facial signal, explicit training is likely necessary (Correia-Caeiro et al., 2023), as adults also performed poorly at recognizing and labeling dog facial signals (Correia-Caeiro et al., 2020). It is likely that like adults, children may also find it difficult to recognize the nonaffiliative facial signals of cats, which means early intervention is essential for correcting misunderstandings.

Study Limitations

Our study is among the first to use an experimental paradigm to assess guardian perceptions of intraspecific cat facial signals and predict subsequent behaviors. However, there are several limitations. First, we utilized video footage of nonfamiliar cats interacting socially in a unique setting: a cat café located in Los Angeles, California. It would be advantageous to have video footage from a broader and more diverse sample of cats, particularly from populations of cats native to the guardians' area (in our study, New Zealand). It would also be helpful to have video footage of cats interacting in various settings and from different angles. Some of our participants mentioned having difficulty identifying and perceiving facial signals due to the type of video footage they were presented with. Second, our study only asked guardians to evaluate 10 facial signals (5 affiliative and 5 nonaffiliative) and associative clips. Given that cats can produce a wide

and diverse array of facial signals (276 in total; Scott & Florkiewicz, 2023), it is essential for follow-up studies to include video footage that features a broader range of morphologically distinct facial signals. Moreover, our video footage included other facial muscle movements, such as eyelid movements, which do not strongly correlate with either context. Therefore, it is important to understand how the addition of these movements influences predictive accuracy. This idea raises a third point: Future studies should focus on evaluating how accurately guardians can perceive their own cats' play faces and predict various play outcomes. This will help confirm whether there are challenges in recognizing nonaffiliative facial signals, which may stem from confusion between intense play sessions and actual fighting. Finally, our participant pool was biased toward older females who owned cats. Previous research has highlighted differences in the ability to accurately interpret cat facial signals based on age and gender (Dawson et al., 2019). Future studies should aim to evaluate the predictive abilities of a larger and more diverse demographic (including populations outside of New Zealand).

Conclusion

Our research underscores the complexities of interpreting facial signals and predicting intraspecific social interactions among cats. We emphasize the importance of developing specific educational tools for cat guardians and young children, created by professionals with experience in both research and applied settings. By helping guardians understand the key differences in facial muscle movements associated with affiliative and nonaffiliative facial signals, such as differences in ear movements, they can enhance their observational skills and contribute positively to the welfare of cats.

Summary for Practitioners

The number of multicat households and shelters has increased in recent years, creating more opportu-

nities for social interactions among cats in indoor environments. Although extensive research exists on cat-human interactions, there are relatively few studies examining cat-cat interactions, and even fewer explore how well humans can accurately perceive and predict the outcomes of these interactions using facial signaling information. Understanding how humans perceive the facial signals and associated interactions between cats is crucial, as conflicts among cats can cause significant stress for their guardians. Conflict between cats can complicate the efforts of caregivers in shelters and clinical settings, making it difficult to provide optimal care for cats in group environments. We distributed a survey to individuals ($N = 598$) subscribed to the Companion Animals New Zealand newsletter to assess cat guardians' effectiveness in perceiving and predicting various social interactions among cats. Our survey consisted of four parts: (1) assessing participants' experiences and knowledge of cats; (2) an experimental video paradigm; (3) a follow-up assessment; and (4) a demographic questionnaire. For the experimental paradigm, participants viewed 10 video clips (five affiliative and five nonaffiliative) showcasing the beginning of cat interactions and were asked to predict the outcomes based on facial signals. Specifically, participants had to determine whether the outcome would be friendly or not and describe their predictions. Afterward, they watched complete interactions to revise their responses. A summary of our key findings and suggestions for practitioners is presented below.

Importance of Training Tools

When we asked participants to assess their confidence in reading body language (including facial signals) and predicting the behaviors of cats, companion animals, livestock, and wildlife, we found that their confidence decreased after completing our survey. This decline occurred despite the fact that our survey focused solely on the facial signals of unfamiliar domesticated cats. Our results suggest that generally speaking, training tools would be beneficial for

animal guardians. By boosting their confidence in understanding body language and predicting animal behaviors, these tools could enhance guardians' skills in resolving conflicts when animals clash. Additionally, they may lead to more positive interactions between humans and animals.

Challenges with Nonaffiliative Social Interactions

In our study, we discovered that participants had the most difficulty accurately predicting when a nonaffiliative social interaction would occur based solely on facial signals. However, this difficulty was limited to making predictions. Generally, participants were able to adjust their responses correctly regarding nonaffiliative social interactions after watching the entire interaction. During our analysis of open-ended responses, we discovered that much of the confusion likely stems from participants' difficulty in distinguishing signals of rough play from aggression. When developing tools to help guardians accurately interpret the communicative signals of cats and predict the outcomes of their interactions, it's important to focus on the subtle differences between playful expressions and those that indicate aggression or defense. Emphasizing ear and mouth movements can enhance this understanding.

Professional Experience Matters

While most participants struggled to predict nonaffiliative interactions between cats based on facial signals, those with professional experience with animals performed better than those without such experience. This result indicates that practitioners are in a unique and important position to provide guardians with accurate information about the communicative abilities of cats. They should be at the forefront of developing training tools. We suggest that these training tools be tailored not only to adults but also to young children, as previous research has found that young children struggle to accurately identify facial signals of domesticated animals like cats.

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